LESSON 7.1

Explore Integers and Their Opposites

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use integers to represent quantities in everyday life, explain what 0 means in each situation, and name the opposite of an integer on the number line.

WHY THIS MATTERS The lowest elevation in North America is Badwater Basin in Death Valley: 86 meters below sea level. When we measure elevation, sea level is defined as 0. So 86 meters below sea level cannot be written as plain 86 — that would put it ABOVE the water. Numbers less than zero are negative numbers, written with a − sign, so the elevation of Badwater Basin is −86 meters. On a number line, −86 is 86 units to the LEFT of 0, while 86 is 86 units to the RIGHT of 0. Those two numbers are opposites.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Integer *Spanish: Número entero*	The counting numbers, their opposites, and 0.	−9, −1, 0, 6, 230
Opposite *Spanish: Opuesto*	A number that is the same distance from 0 as another number, but on the other side of 0.	−6 and 6 are opposites
Negative number *Spanish: Número negativo*	A number less than 0, written with a − sign. On a number line it sits to the left of 0.	−9 is read negative nine
Positive number *Spanish: Número positivo*	A number greater than 0. On a number line it sits to the right of 0.	6, 19, 230
Number line *Spanish: Recta numérica*	A straight line where numbers are placed in order. It extends to the left of 0 to show values less than 0.	left is less, right is more
Opposite of the opposite *Spanish: Opuesto del opuesto*	Taking the opposite twice brings you back to the first number.	−(−6) = 6

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A game score starts at 0. You earn 8 points, then you lose 8 points. Write each as an integer.

1. 0 means the starting score.
2. Earn 8 points: 8. Lose 8 points: −8.
3. Both are 8 units from 0, on opposite sides.

Answer: 8 and −8 are opposites

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

One day in January the temperature in Toronto was 9°C and in Reykjavik it was −9°C.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The integer _____ represents _____ .
- Zero stands for _____ in this situation.
- The opposite of _____ is _____ because _____ .
- On the number line, _____ sits _____ units to the _____ of 0.

Word bank integer · opposite · negative · positive · number line · zero · distance · above · below · left · right · sign

WATCH OUT!

A common mistake

The most common mistake in this lesson is thinking that the opposite of a negative integer is still negative

LESSON 7.1

Explore Integers and Their Opposites

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Write the integer that represents each situation. Remember to decide what 0 means first.

Situation	Integer
A football play with a 12-yard loss	
A hiker 250 meters above sea level	
7 degrees below zero	
A submarine resting at the surface of the water	

HINT Each row describes a quantity measured from a starting point. That starting point is 0.

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each integer by which side of 0 it sits on when you graph it on a number line.

Groups: **Left of 0 (negative integer)** **Right of 0 (positive integer)**

_____ -86

_____ 9

_____ -9

_____ 21

_____ -3

_____ 250

HINT Look only at whether a − sign is in front of the number.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Badwater Basin is 86 meters below sea level, and sea level is 0. Which integer represents that elevation?

(A) -86

(B) -68

(C) 0

(D) 86

HINT Sea level is 0. Is 'below sea level' more than 0 or less than 0?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

In Reykjavik the temperature was -9°C. What is the opposite of -9?

(A) -18

(B) -9

(C) 0

(D) 9

HINT 'Opposite' means the same distance from 0, but on the other side of 0.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is the opposite of 0?

(A) A number written -0, which is less than 0

(B) Every negative number

(C) Zero has no opposite

(D) 0

HINT The opposite of a number is the same distance from 0 on the other side.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake: The Opposite of -7

1. **Problem** Name the opposite of -7.
2. **Student thinking** The number starts out negative, and taking the opposite of a negative number keeps it negative, so the answer stays on the left side of 0.
3. **Student answer** The opposite of -7 is -7.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT What does the word 'opposite' require about which side of 0 a number is on?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide what 0 meant in each situation, and how did that decide the sign of the integer?

SENTENCE STARTER In this situation 0 means ___. The quantity is ___ zero, so I represent it with a ___ integer, which is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 7.1

Explore Integers and Their Opposites

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Write the opposite of each integer.

Integer	Opposite
-15	
8	
0	
-100	

SHOW YOUR WORK

2 **Mark and label each value on the number line.**

Graph these integers: -9, 9, 0, -4, 6



SHOW YOUR THINKING

3 **Circle the letter of the best answer. Show how you know.**

Which statement about 0 is TRUE?

(A) 0 is an integer, and it is its own opposite

(B) 0 is a negative integer because it is not positive

(C) 0 is a positive integer because it has no negative sign

(D) 0 is not an integer at all

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Which list of integers is ordered from LEAST to GREATEST?

(A) -8, -3, 0, 5

(B) -3, -8, 0, 5

(C) 0, -3, -8, 5

(D) 5, 0, -3, -8

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

Badwater Basin is at -86 meters. Which elevation is FARTHER below sea level than Badwater Basin?

(A) -120 meters
(B) -40 meters
(C) 0 meters
(D) 86 meters

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Mistake: Reykjavik's Temperature Swing

1. **Problem** At 6 a.m. in Reykjavik the temperature was -12°C. At 2 p.m. it was the opposite of the 6 a.m. temperature. What was the temperature at 2 p.m.?
2. **Student thinking** The opposite of a temperature should still be a cold temperature, so I keep the negative sign and just make the number bigger. That gives a temperature farther from 0.
3. **Student answer** The temperature at 2 p.m. was -24°C.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did you decide what 0 meant in each situation, and how did that decide the sign of the integer?

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LESSON 7.1

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Order these integers from LEAST to GREATEST.

Groups:

_____ 9

_____ -86

_____ 0

_____ -9

_____ 21

_____ -3

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

At 6 a.m. in Reykjavik the temperature was -12°C. At 2 p.m. it was the opposite of the 6 a.m. temperature, and at 11 p.m. it was the opposite of the 2 p.m. temperature. What were the temperatures at 2 p.m. and 11 p.m.?

(A) -12°C and 12°C

(B) 0°C and 12°C

(C) 12°C and -12°C

(D) 12°C and 24°C

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A football team starts at the line of scrimmage, which is 0. During one series they had a 10-yard loss, a 4-yard gain, a 3-yard loss, and a 21-yard gain. Where does the series leave them?

(A) 8 yards past the line of scrimmage

(B) 12 yards past the line of scrimmage

(C) 18 yards past the line of scrimmage

(D) 38 yards past the line of scrimmage

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Describe two different real-world situations that use negative integers. For each one, tell exactly what 0 means in that situation, write one negative integer it could produce, and explain what the opposite of that integer would mean in the same situation.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How did you decide what 0 meant in each situation, and how did that decide the sign of the integer?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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